Case-Study

Paris Olympic Games 2024 (Part A)

INDEX

BUSINESS STORY

Paris is using state-of-the-art technology to transform event management as it prepares to host the Summer Olympics in 2024. The Event Management Platform (EMP), an advanced technology is created to smoothly coordinate every facet of the Olympic Games, is at the centre of this undertaking.

The EMP acts as the hub for planning and carrying out a wide range of events held in famous locations throughout Paris. The EMP guarantees perfect coordination at every stage, from the thrilling opening ceremony at Paris Olympic Stadium to the suspenseful competitions at Paris International Aquatic Centre.

The effective distribution of athlete accommodations is one of the main features of the EMP. The technology ensures comfort and convenience for athletes from across the world by maximizing housing arrangements based on variables like team size, preferences, and proximity to event venues.
Furthermore, the EMP carefully monitors and controls inventory of sporting goods, guaranteeing that all equipment is in excellent shape & easily accessible when needed. This strategy increases game efficiency and reduces downtime.

An app for smartphones that is specifically designed to improve the viewing experience is smoothly integrated with the EMP. In addition to purchasing tickets, spectators can get real-time schedule updates, and navigate the city with ease, thanks to the app's intuitive interface and advanced features.

The EMP protects sensitive data and guarantees adherence to international standards by implementing strong security protocols. This dedication to safety is a safe and pleasurable atmosphere for all parties concerned by instilling confidence in subjects, spectators, and stakeholders alike.

The EMP will match the events with the venues to avoid any discrepancies of overlapping of matches and athlete’s schedule. Athletes get their accommodation, matches to play and venues to play those game on as well as events take place on right venues with no overlapping and at the same time as volunteer reaches the event he has been rostered.

EMP will also take care of inventory count of equipment matching with the venues.
The EMP will leave an mark and raise the standard for organizing events worldwide when the 2024 Olympics come to an its conclusion

Identifying the Nouns

Paris: The city mentioned as the host for the Summer Olympics in 2028.

Accommodation - Manages lodging details for athletes, considering dynamic adjustments based on team sizes and other requirements.

State-of-the-art technology: A term used to describe the latest and most advanced level of technology available.

Event management: The process of planning, organizing, and executing events.

Summer Olympics: A major international multi-sport event held every four years.

Equipment - Manages inventories and maintenance schedules of sports gear and other essential resources

Venue - Represents each location across Paris where Olympic events will be held.

Event Management Platform (EMP): The advanced technology system mentioned as being used to coordinate the various aspects of the Olympic Games.

Olympic Games: An international sporting event divided into summer and winter sports where thousands of athletes from around the world participate in a variety of competitions.

Paris: The city referred to as the location with famous venues for the events.

Paris Olympic Stadium: A specific venue mentioned for the opening ceremony..

Athlete accommodations: Housing provided for athletes participating in the Olympic Games.

Team size, preferences, proximity: Variables considered for housing arrangements in athlete accommodations.

Inventory of sporting goods: The stock of sports equipment needed for the Olympic Games.

App for smartphones: A mobile application designed to enhance the spectator experience at the Olympics.

Tickets, schedule updates, city navigation: Features and functionalities of the smartphone app.

Security protocols: Procedures and measures implemented to protect sensitive data and ensure safety.

Athlete - Manages details for each participating athlete, including nationality, team affiliation, and specific needs.

Volunteer - For managing details of each volunteer, such as skills, availability, and assignments.

International standards: Official requirements that need to be met to maintain quality and safety.

Spectators: People who watch the Olympic Games, either in person or through media.

Stakeholders: Individuals or organizations with an interest in the Olympic Games, such as sponsors, broadcasters, and sports federations.

Event - Handles scheduling and details of various Olympic events, including the opening ceremony

Legacy: The long-lasting impact of the Olympic Games on the host city.

Identifying Entities

1. **Athlete** - Manages details for each participating athlete, including nationality, team affiliation, and specific needs.
2. **Volunteer** - For managing details of each volunteer, such as skills, availability, and assignments.
3. **Event** - Handles scheduling and details of various Olympic events, including the opening ceremony.
4. **Venue** - Represents each location across Paris where Olympic events will be held.
5. **Accommodation** - Manages lodging details for athletes, considering dynamic adjustments based on team sizes and other ments.
6. **Equipment** - Manages inventories and maintenance schedules of sports gear and other essential resources.

Identifying Relationships

- **Event to Venue:**
  An Event is held at one Venue.
  A Venue can host many Events.
- **Event to Equipment:**
  An Event may use many pieces of Equipment.
  Equipment can be used in one or many Events.
- **Athlete to Accommodation:**
  An Athlete stays in one Accommodation.
  An Accommodation can house many Athletes.
- **Athlete to Event:**
  An Athlete can participate in one or many Events.
  An Event can have many Athletes participating.
- **Volunteer to Event:**
  A Volunteer can be assigned to one or many Events.
  An Event can have many Volunteers assigned.

Describe relationships.

I. **Event host Venue:**

Event to Venue:

Cardinality: Each Event is held at exactly one Venue (1).

Participation: Each Event must be held in a Venue (Mandatory).

Type: Classifying relationship (Events are categorized by their location).

Venue to Event:

Cardinality: A Venue can host zero to many Events (0...*).

Participation: A Venue may or may not host any Events (Optional).

Type: Transactional relationship (Venues facilitate various events).

So, Overall Event hosts Venue in a 1 to Many relation.

II. **Event require Equipment:**

Event to Equipment:

Cardinality: An Event may use zero to many pieces of Equipment (0...*).

Participation: Using Equipment is optional for an Event.

Type: Transactional relationship (Equipment use is part of event execution).

Equipment to Event:

Cardinality: Equipment can be used in zero to many Events (0...*).

Participation: Equipment may or may not be used in any Event (Optional).

Type: Transactional relationship (Equipment is utilized across events).

So, Overall Event hosts Venue in a Many to Many relation. With associate entity

III. **Athlete stays in Accommodation:**

Athlete to Accommodation:

Cardinality: Each Athlete stays in exactly one Accommodation (1).

Participation: Every Athlete must have an Accommodation (Mandatory).

Type: Transactional relationship (Accommodations are part of athlete management).

Accommodation to Athlete:

Cardinality: An Accommodation can house zero to many Athletes (0...*).

Participation: An Accommodation may or may not house any Athletes (Optional).

Type: Transactional relationship (Accommodations serve athletes).

So, Overall Event hosts Venue in a 1 to Many relation

IV. Athlete participates Event:

Athlete to Event:

Cardinality: An Athlete can participate in one to many Events (1...*).

Participation: Each Athlete must participate in at least one Event (Mandatory).

Type: Transactional relationship (Participation in events defines athlete involvement).

Event to Athlete:

Cardinality: An Event can have zero to many Athletes (0...*).

Participation: An Event may or may not have any Athletes (Optional).

Type: Transactional relationship (Events are defined by athlete participation).

So, Overall Event hosts Venue in a Many to Many relation. With associate entity

V. Volunteer deployed Event:

Volunteer to Event:

Cardinality: A Volunteer can be assigned to one to many Events (1...*).

Participation: Each Volunteer must be assigned to at least one Event (Mandatory).

Type: Transactional relationship (Assignments define volunteer activity).

Event to Volunteer:

Cardinality: An Event can have zero to many Volunteers (0...*).

Participation: An Event may or may not have any Volunteers (Optional).

Type: Transactional relationship (Events require volunteer support).

So, Overall Event hosts Venue in a Many to Many relation. With associate entity

The Table -:

The Entity’s Name, Attribute, Description, Data Type, Key type

Name	Attribute	Description	Data Type	Key Type
Venue	Venue_ID	ID for each venue location	INT	Primary Key
	Venue_Name	Name of the venue	VARCHAR(50)	
	Location	Location of the venue in Paris	VARCHAR(50)	
	Capacity	Maximum number of people to accommodate	VARCHAR(50)	
Event	Event_ID	Unique identifier for each event	INT	Primary Key
	Event_Name	Name of the event	VARCHAR(50)	
	Event_Type	Type or category of the sport	VARCHAR(50)	
	Start_Time	Time when the event will start	datetime	
	End_Time	Time when the event will end	datetime	
	Athlete_ID	Unique identifier for each athlete	INT	Foreign Key
	Venue_ID	ID of the venue where the event takes place	INT	Foreign Key
Athlete	Athlete_ID	Unique identifier for each athlete	INT	Primary Key
	Athlete_Name	Name of the athlete	VARCHAR(50)	
	Nationality	Country the athlete represents	VARCHAR(50)	
	Sport	Sport the athlete competes in	VARCHAR(50)	
	Gender	Gender of the athlete	VARCHAR(50)	
	Accommodation_ID	ID of the athlete's accommodation	INT	Foreign Key
Equipment	Equipment_ID	Unique identifier for each piece of equipment	INT	Primary Key
	Equipment_Type	Type of equipment	VARCHAR(50)	
	Sport	Sport associated with the equipment	VARCHAR(50)	
	Event_ID	Identifies the equipment is used in which event	INT	Foreign Key
	Quantity	Quantity of each type of equipment available	INT	
Volunteer	Volunteer_ID	Unique identifier for each volunteer	INT	Primary Key
	Volunteer_Name	Name of the volunteer	VARCHAR(50)	
	Role	Role or job of the volunteer	INT	
	Event_ID	Tells which event the volunteer partici[ated	INT	Foreign Key
	Contact_Info	Contact information for the volunteer	INT	
Accommodation	Accommodation_ID	Unique identifier for each accommodation	INT	Primary Key
	Type	Type of accommodation (e.g., hotel, dormitory)	VARCHAR(50)	
	Location	Location of the accommodation	VARCHAR(50)	
	Capacity	Number of athletes that can be accommodated	VARCHAR(50)	

Normalisation

1. VENUE TABLE

UNF: The table might have multiple columns for events if it were unnormalized. 1NF:

- Venue_ID (Primary Key)
- Venue Name
- Location
- Capacity

2NF: The table is already in 2NF because all non-key attributes are fully functionally dependent on the primary key.

3NF: The table is already in 3NF because there are no transitive dependencies.

2. Event Table

UNF: The table might have multiple columns for athletes and equipment if it were unnormalized. 1NF:

- Event_ID (Primary Key)
- Event_Name
- Event_Type
- Start_Time
- End_Time

2NF:

- Venue_ID (Foreign Key) introduced to link event with venues.
- Same as 1NF because all non-key attributes are fully dependent on the primary key.

3NF:

- Same as 2NF because there are no transitive dependencies.

3. Athlete Table

UNF: The table might contain details of events, accommodations, and medals if it were unnormalized. 1NF:

- Athlete_ID (Primary Key)
- Athlete_Name
- Nationality
- Sport
- Gender

2NF:

Accommodation_ID (Foreign Key) Now setting this up for Foreign key

Same as 1NF because all non-key attributes are fully dependent on the primary key.

3NF:

Same as 2NF because there are no transitive dependencies.

4. Equipment Table

UNF: There could be details about which events use each piece of equipment if it were unnormalized. 1NF:

- Equipment_ID (Primary Key)
- Equipment_Type
- Sport
- Quantity

2NF:

Same as 1NF because all non-key attributes are fully dependent on the primary key.

3NF:

Same as 2NF because there are no transitive dependencies.

5. Volunteer Table

UNF: Might include columns for each event the volunteer has worked at if unnormalized. 1NF:

- Volunteer_ID (Primary Key)
- Volunteer_Name
- Role
- Contact_Info

2NF:

Event_ID (Foreign Key) tells us for the connection between event and volunteer.

Same as 1NF because all non-key attributes are fully dependent on the primary key.

3NF:

Same as 2NF because there are no transitive dependencies.

6. Accommodation Table

UNF: Might include columns for each athlete staying if unnormalized. 1NF:

- Accommodation_ID (Primary Key)
- Type
- Location
- Capacity

2NF:

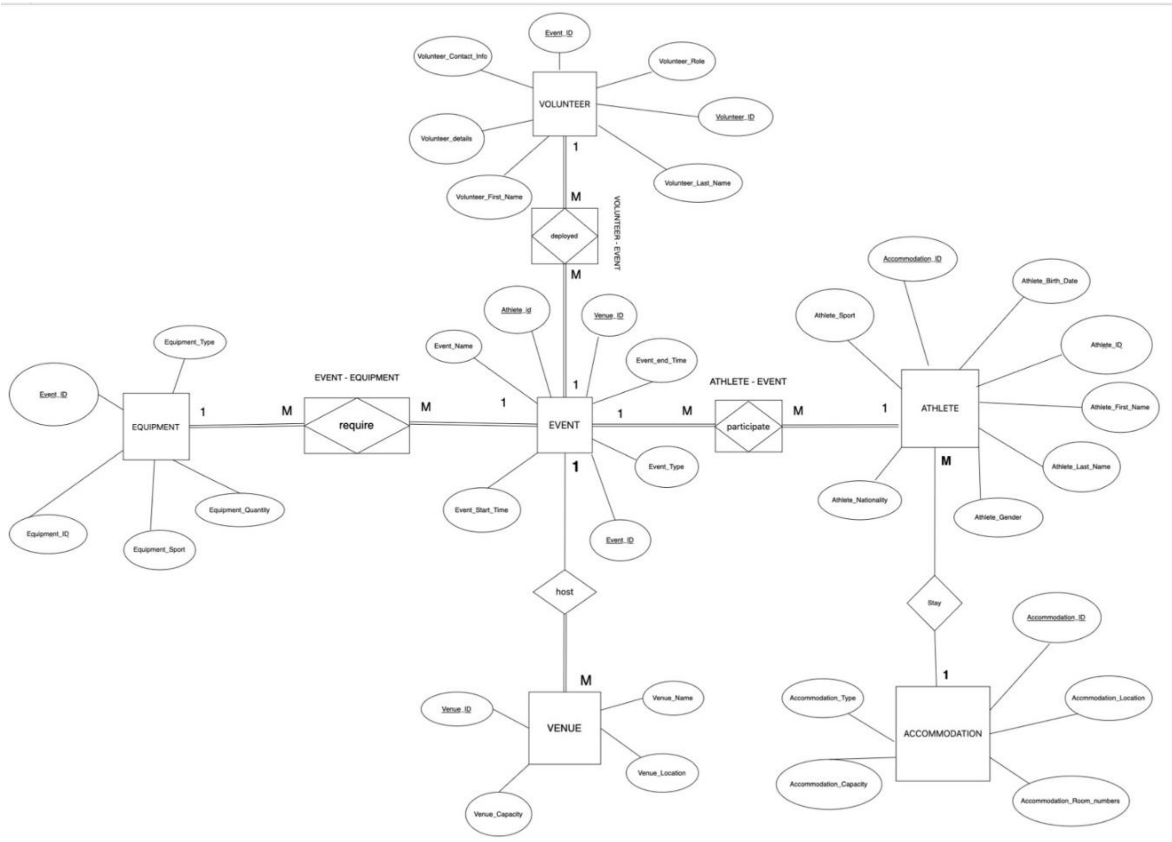
Same as 1NF because all non-key attributes are fully dependent on the primary key.

3NF:

Same as 2NF because there are no transitive dependencies.

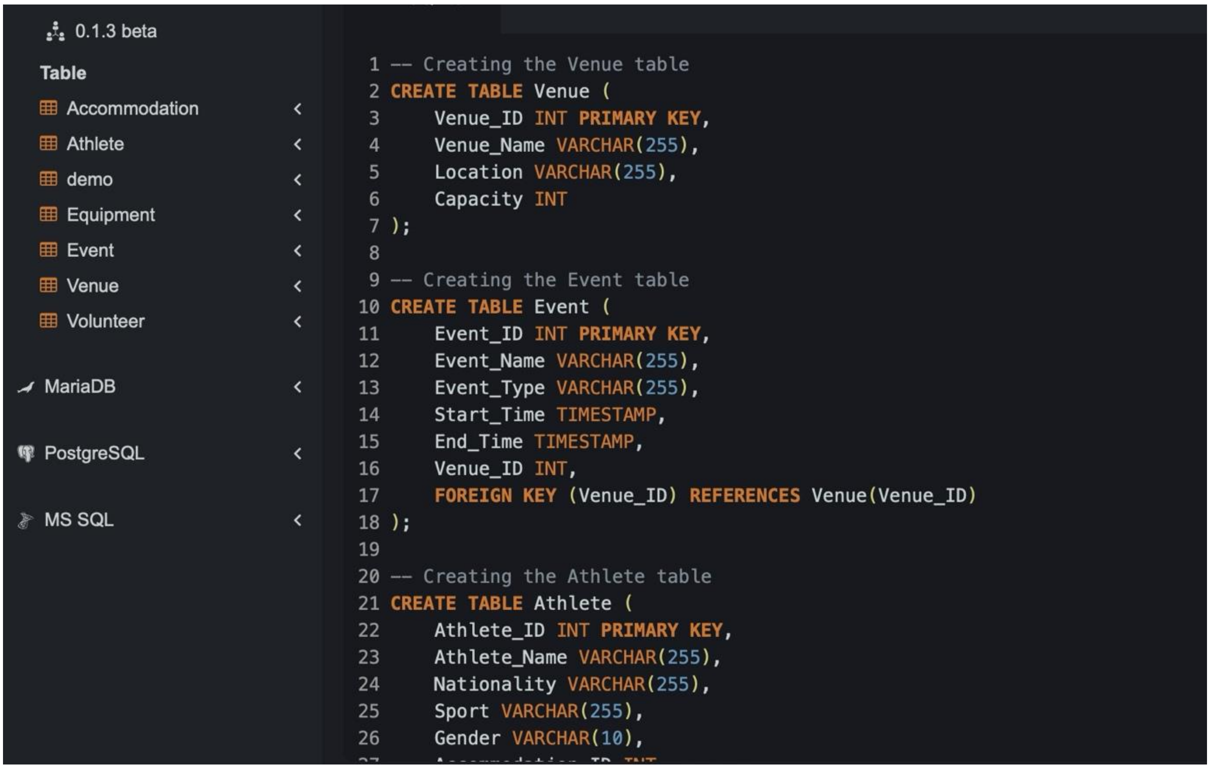
Each of these tables in their current form already adheres to the rules of 3NF based on the attributes provided. There is a clear separation of concerns with primary keys serving as the sole identifiers for records in each table and foreign keys appropriately linking related records across tables. If there were derived attributes or if any non-key attributes depended on other non-key attributes, those would have to be refactored into their own tables to maintain 3NF integrity.

ERD DIAGRAM AS PER CHEN'S NOTATION

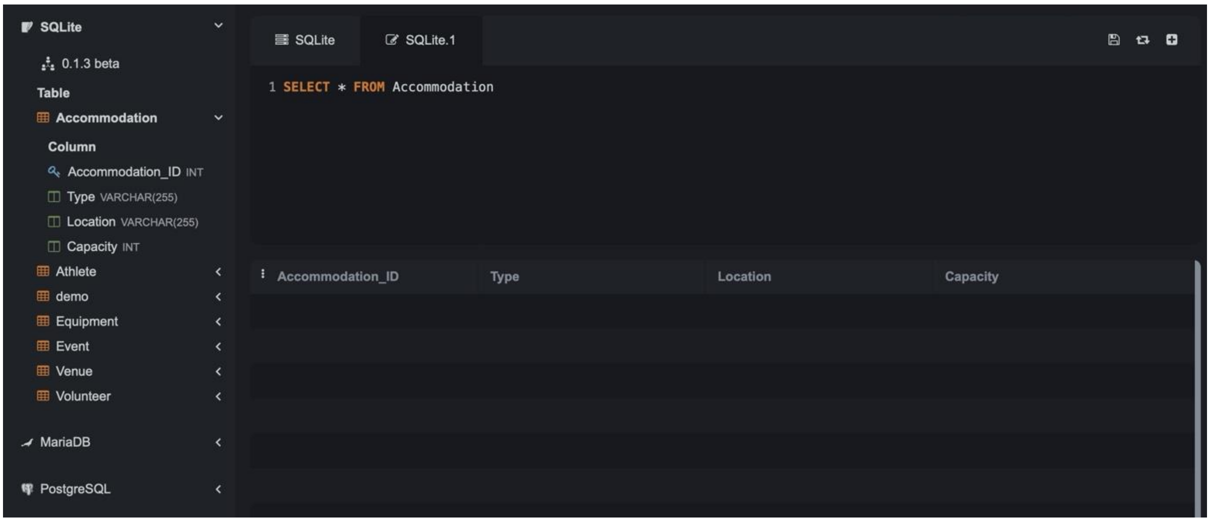


3.0 Create the Database

Create the tables –
Accommodation, Athlete, Events, Equipment, Venue and Volunteer



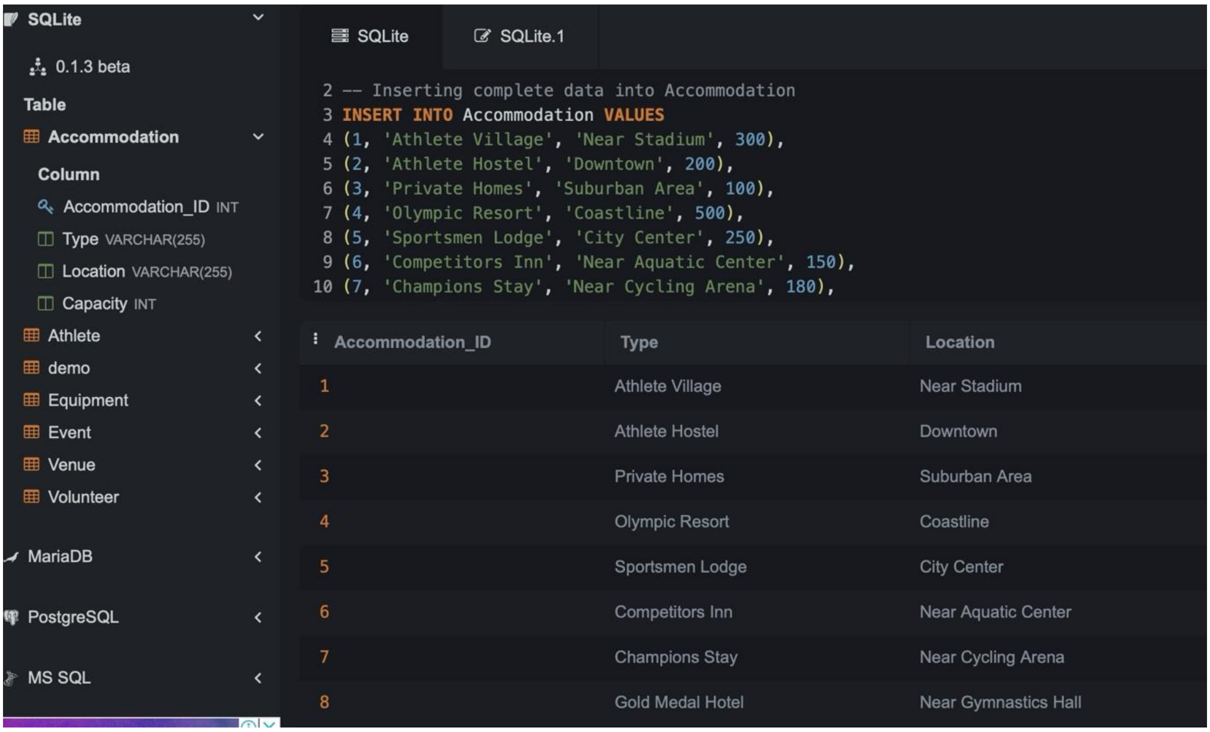
Here's the snippet of the Empty Accommodation table



INSERTING DUMMY DATA ALONGWITH OUTPUT

Table Accommodation

I have inserted the rows of dummy data in the accommodation table.




```sql
-- Inserting complete data into Accommodation
INSERT INTO Accommodation VALUES
(1, 'Athlete Village', 'Near Stadium', 300),
(2, 'Athlete Hostel', 'Downtown', 200),
(3, 'Private Homes', 'Suburban Area', 100),
(4, 'Olympic Resort', 'Coastline', 500),
(5, 'Sportsmen Lodge', 'City Center', 250),
(6, 'Competitors Inn', 'Near Aquatic Center', 150),
(7, 'Champions Stay', 'Near Cycling Arena', 180),
```

Accommodation_ID	Type	Location
1	Athlete Village	Near Stadium
2	Athlete Hostel	Downtown
3	Private Homes	Suburban Area
4	Olympic Resort	Coastline
5	Sportsmen Lodge	City Center
6	Competitors Inn	Near Aquatic Center
7	Champions Stay	Near Cycling Arena
8	Gold Medal Hotel	Near Gymnastics Hall

MACQUARIE
University

Individual Assignment – Case Study Information

Paris Olympic Games 2024 (Part B)

Insights and Recommendations

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Submission Date	:	29 APRIL 2024

Creating Views to report the findings

SQL QUERIES WITH RESULTS:

1. Number of Events taking place in the venue

The rationale of the query is to know how the venues are being effectively used. The idea of the project will be to check if venues are under or overutilized. Underutilization speaks to the concept that a venue's potential is not completely exhausted, while overutilization may damage the facilities of a venue through too much activity, or otherwise render the facilities unusable, through crowding, or overusing its resources.

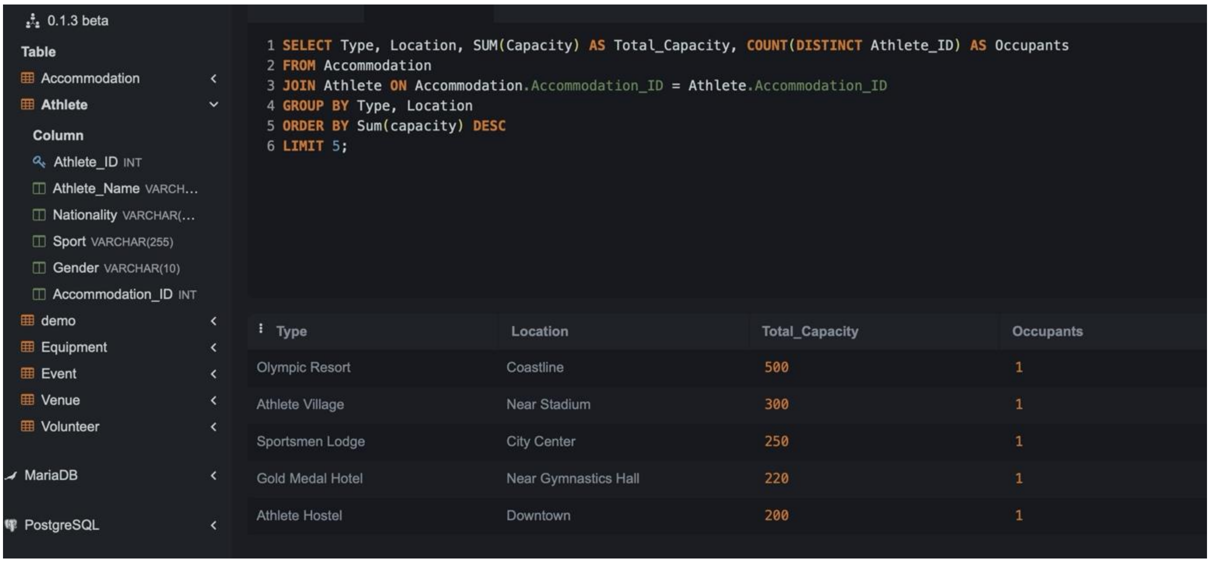
The implication is that from the query outcome, with a clear understanding of the uses of venue, the organizers will make well-versed decisions about places to hold events. For example:

- If overuse of the facilities is a finding at one venue, events could be redistributed to other venues in order to share the load, with potential benefits for the experience of the attendee and extended life of the facilities. If done otherwise, the organizers may end up overscheduling events in a given venue to maximize the return of their facilities investment. To this end, both can help reduce operational costs; for instance, consolidating events at fewer venues could mean less spending on staff, security, maintenance, and energy. This would result in efficiency, sustainability, and cost-effectiveness in the organization of events.

2. Usage of the Equipment

1. **Maintenance Prioritization**: Knowing which pieces of equipment are used most often means that these items can be given special attention in terms of maintenance. Regular and proactive maintenance on these frequently used items can prevent unexpected failures that might disrupt events, ensuring reliability and consistent performance. This helps in reducing downtime and potentially costly repairs that occur due to equipment failure during critical times.
2. **Strategic Procurement**: Understanding which equipment is in high demand provides valuable insight for procurement strategies. The person can use this information to make informed decisions about purchasing or leasing equipment. For instance, if certain equipment is constantly in use, it might be more cost-effective in the long term to purchase rather than lease. Additionally, this data supports making bulk purchases, which often come with discounts. Bulk purchasing not only reduces the per-unit cost but might also ensure that there is always backup equipment available when needed.
3. **Negotiating Leasing Terms and Discounts**: With concrete data on which equipment types are most essential, the person can enter negotiations with suppliers from a position of strength. They can negotiate better leasing terms or discounts based on bulk purchases. This might include lower rates, better maintenance terms, or favorable leasing conditions, which can significantly reduce overall costs.

3. Accommodation Analysis of the Athletes




```sql
SELECT Type, Location, SUM(Capacity) AS Total_Capacity, COUNT(DISTINCT Athlete_ID) AS Occupants
FROM Accommodation
JOIN Athlete ON Accommodation.Accommodation_ID = Athlete.Accommodation_ID
GROUP BY Type, Location
ORDER BY Sum(capacity) DESC
LIMIT 5;
```

Type	Location	Total_Capacity	Occupants
Olympic Resort	Coastline	500	1
Athlete Village	Near Stadium	300	1
Sportsmen Lodge	City Center	250	1
Gold Medal Hotel	Near Gymnastics Hall	220	1
Athlete Hostel	Downtown	200	1

The reason to monitor and keep records of the use efficiency in the different types of accommodations for athletes lies in knowing the capacity of the facilities (maximum occupancy) and the number of occupants present at the moment. With these two metrics in clear view, it assists administrators and management in better making a judgement call on whether the accommodations are being effectively used or if indeed there are inefficiencies existing.

Capacity vs. Occupational Analysis

Capacity: This is the maximum number of human beings that a given accommodation can host. It is either pegged by space, security considerations, or the spatial design of a facility.

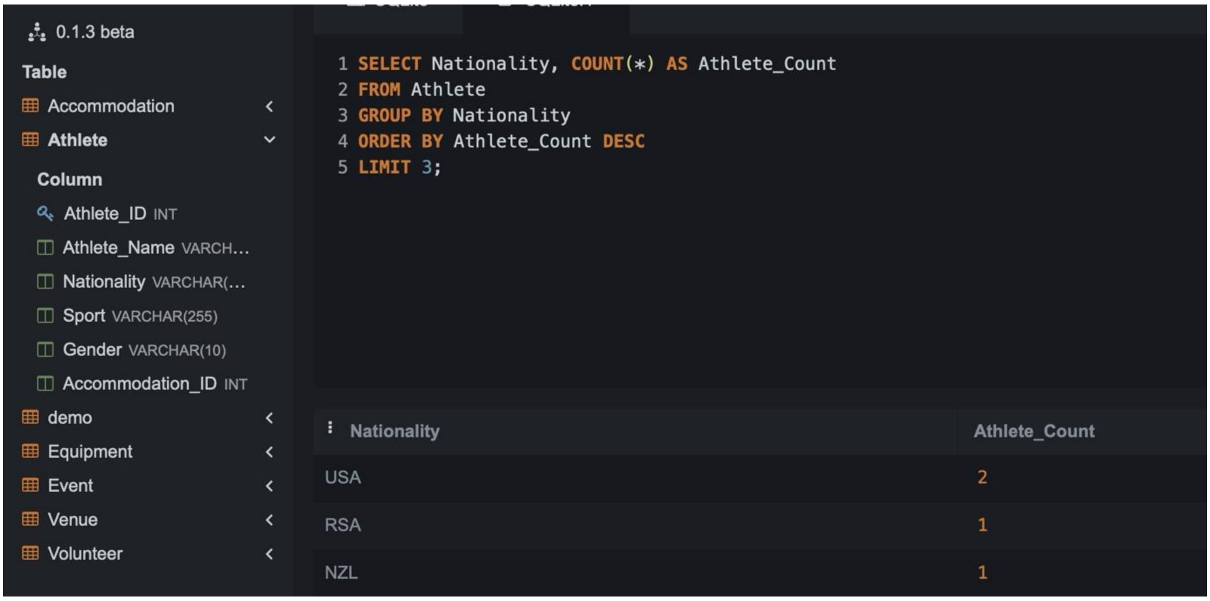
Current Occupancy: This actually refers to the actual number of occupants who are actually staying within the accommodation at any one given time.

This will provide information with which the management can determine the rate at which the current utilization of space compares to the capacity. This is an important step to ensure planning and operational efficiency.

Definition: When more residents than the accommodation capacity are staying.

Implications: The implications of this overcapacity can be many, like reduced comfort for consumers, safety risks, and in fact, violation of law or regulation standard. On the flip side, it may point out to either a condition of misestimate space requirements or a situation where the company is going through a temporary high.

4. Athlete's Count on Nationality



Tracking the participation levels of athletes from different countries in various sports and competitions can provide valuable insights for several stakeholders in the sports industry. Here's a detailed look at how this data can be leveraged and its implications:

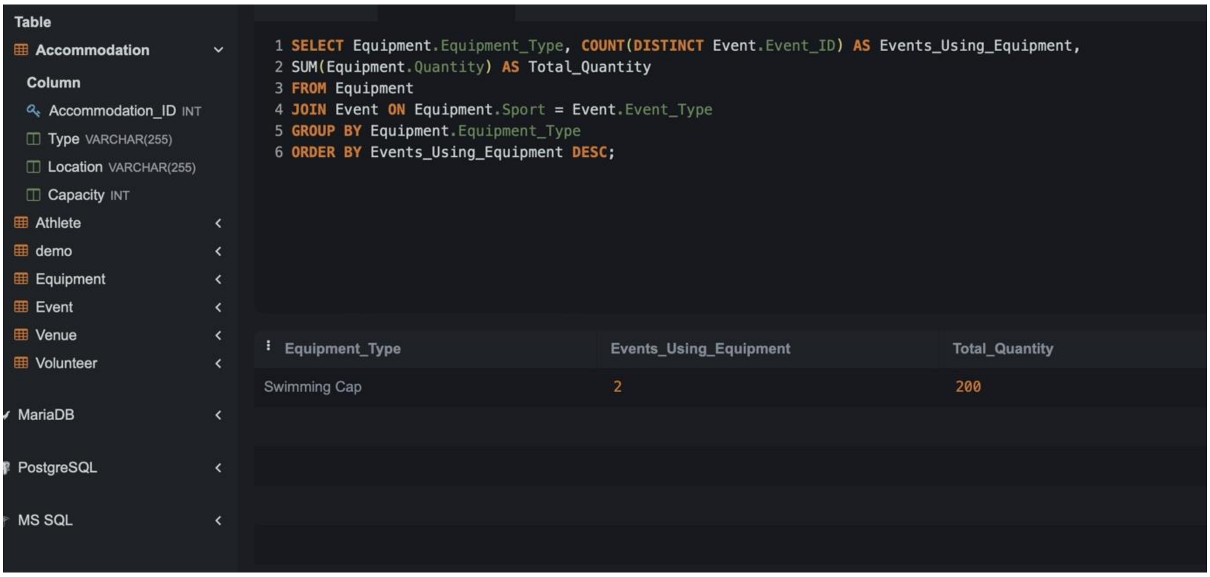
1. Marketing Strategies

- **Targeted Advertising**: Knowing which countries have high levels of participation can help marketers tailor their advertising campaigns to specific audiences. For example, a country with a high number of soccer players might be a prime target for soccer gear advertisements.
- **Product Development**: Understanding the sports that are popular in different regions can guide companies in developing specialized products that meet the specific needs of athletes in those sports.

2. Promotional Strategies

- **Event Planning**: By analyzing athlete participation data, event organizers can choose locations that are likely to attract more participants and spectators, maximizing attendance and engagement.
- **Localized Promotions**: Promotions can be customized based on the popularity of a sport in a particular area. For instance, offering discounts on basketball merchandise in a country where basketball is more popular can increase sales.

5. Inventory Count on Equipment's



o delve deeper into this topic, let's break down the analysis and implications of monitoring and optimizing equipment usage across different events.

Explanation of the Analysis:

1. **Data Collection**: The first step involves gathering comprehensive data on the types of equipment used at various events. This data collection would cover details such as the quantity of each item used, duration of usage, and the type of event (e.g., corporate meetings, weddings, concerts).
2. **Equipment Categorization**: After collecting the data, the next step is to categorize the equipment based on its function and frequency of use. For instance, audio-visual equipment like projectors and microphones, seating arrangements like chairs and tables, and specialized equipment such as lighting rigs or stage setups.

Recommendations-:

Streamline Volunteer Management: Strengthen the scheduling system with the objective to optimally deploy volunteers based on event demands and their availability. This will reduce fatigue and improve efficiency, and also make sure that all areas are well-covered and not overstaffed.

Improved Spectator Experience: The data from historical events will be used to help in the proper management of crowds and improving the experience of spectators. The notification should also be issued by the mobile apps on the timing of events, seating capacity, and other facilities. This would add to the convenience of spectators and hence increase satisfaction, which may drive higher future attendances.

Improve Transportation Coordination: Analyze local transport use and movement patterns for attendees and aim to come up with the best routes and schedules that will cut down waiting times and congestion. Where possible, sign up local transport providers and issue passes to the event that will reduce travel hassles by attendees and logistic headaches.

Sustainable Practices: Introduce more sustainable practices in operations, from recycling at venues to reducing water and energy usage. This not only does alignment with global environmental goals but also brings sponsorship from companies with green mandates, thus opening new revenue streams.

Digital Engagement: Maximize and make use of all digital platforms to reach out to the global audience, like Virtual Reality experiences of the games, advanced behind-the-scenes content, and others. Games can be more far-reaching and engaging, drawing in audiences that would be unable to attend in person.

Diversify Merchandise Offerings: According to sales data, expand and diversify merchandise offerings in a way that limited edition products could sometimes be created, which would create a buzz and attract more buyers, boosting the overall merchandise revenue.

Innovations in Health and Safety: With world current health conditions, further innovate in health safety applications at venues. These may include contactless systems, real-time health monitoring, and rapid response protocols for anyone present in the venue.

Flexible Pricing of the Accommodation: Apply flexible pricing mechanisms to the accommodations that the athletes will not utilize in totality. Sell them at throwaway prices to any last-minute booking from an employee or a traveller, and in this way, they make revenue from this type of service.

Event Analytics: Build a comprehensive analytics framework that keeps tracking event performance in real time constantly. Insights from this exercise will be used to make on-the-spot adjustments of events on-the-fly, for example, rescheduling events, relocating them to different places, or realigning with actual demand resources.

Communication with stakeholders, including athletes, sponsors, media, and local authorities, should be improved. There are updates from time to time, and transparent communication can improve coordination and satisfaction, hence the realization of the Games' objectives.

Submitted By,

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